Project Report Format

1. **INTRODUCTION**
    1.1 Project Overview
    1.2 Purpose
2. **IDEATION PHASE**
    2.1 Problem Statement
    2.2 Empathy Map Canvas
    2.3 Brainstorming
3. **REQUIREMENT ANALYSIS**
    3.1 Customer Journey map
    3.2 Solution Requirement
    3.3 Data Flow Diagram
    3.4 Technology Stack
4. **PROJECT DESIGN**
    4.1 Problem Solution Fit
    4.2 Proposed Solution
    4.3 Solution Architecture
5. **PROJECT PLANNING & SCHEDULING**
    5.1 Project Planning
6. **FUNCTIONAL AND PERFORMANCE TESTING**
    6.1 Performance Testing
7. **RESULTS**
    7.1 Output Screenshots
8. **ADVANTAGES & DISADVANTAGES**
9. **CONCLUSION**
10. **FUTURE SCOPE**
11. **APPENDIX**
    Source Code(if any)
    Dataset Link
    GitHub & Project Demo Link

PLUGGING INTO THE FUTURE: AN EXPLORATION OF ELECTRICITY CONSUMPTION PATTERNS

1. INTRODUCTION

1.1 Project Overview

The **Electricity Consumption Patterns Analysis System** is an interactive analytical solution developed to monitor, evaluate, and visualize localized power distribution and grid stress anomalies across India. By bridging the gap between raw, highly granular time-series load datasets and spatial tracking engines, this platform empowers national grid engineers, energy analysts, and state electricity boards to isolate structural energy anomalies without the need for expensive hardware installations at every substation.

National power grids deal with massive load fluctuations and unpredictable drops that are cognitively difficult to trace in flat spreadsheet models. Mentally processing over 16,800+ lines of raw time-series logs across multiple historical timelines and 5 regional grids is an overwhelming task for infrastructure operators.

This platform transforms those unorganized logs into visual intelligence. It maps daily consumption levels, peak regional draws, and macro-environmental structural shifts (such as the 2020 national lockdown). Using interactive Tableau engines, the system allows users to seamlessly cross-examine 2019 baseline metrics against 2020 disruption trends, allowing planning teams to pinpoint critical deficit margins and predict potential grid overloads instantly.

1.2 Purpose

The primary purpose of this project is to analyze electricity consumption patterns across different states and regions of India and provide meaningful insights through interactive dashboards. The project helps users understand how electricity demand changed before and during the COVID-19 lockdown and supports better decision-making for energy management.

Objectives

• Analyze electricity consumption across different Indian states and regions.

• Compare electricity usage between the years **2019 and 2020**.

• Study the impact of COVID-19 lockdown on electricity demand.

• Develop an interactive Tableau Dashboard for real-time data exploration.

• Create dynamic visualizations such as maps, charts, Top N and Bottom N analysis.

• Enable users to filter data using Year, Month, Region, and State filters.

• Present analytical insights using Tableau Story.

• Support electricity boards, government agencies, researchers, and analysts in making data-driven decisions.

• Publish the dashboard on Tableau Public and integrate it into a Flask web application.

2. IDEATION PHASE

2.1 Problem Statement

Problem Statement (PS)	**I am (Customer)**	**I'm trying to**	**But**	**Because**	**Which makes me feel**
PS-1	Regional Power Grid Operations Manager	Track and isolate weekly energy demand drops across states during the 2020 national lockdown.	Raw time-series data is fields heavy (16,800+ rows) and lags heavily during execution.	Standard spreadsheets and static reports lack flexible, multi-region filter sync options.	Stressed, overwhelmed by unorganized timestamps, and anxious about making incorrect load distribution decisions.
PS-2	National Energy Policy & Demand Analyst	Identify which specific industria states faced the sharpest drops to plan future load scheduling.	Tableau state rankings break or show wrong order when regional filters are changed	Standard filters do not use Context Optimization, mixing up localized state calculations.	Frustrated by technical dashboard glitches, and delayed in presenting accurate baseline vs lockdown insights to heads.

CUSTOMER PROBLEM STATEMENT CANVAS

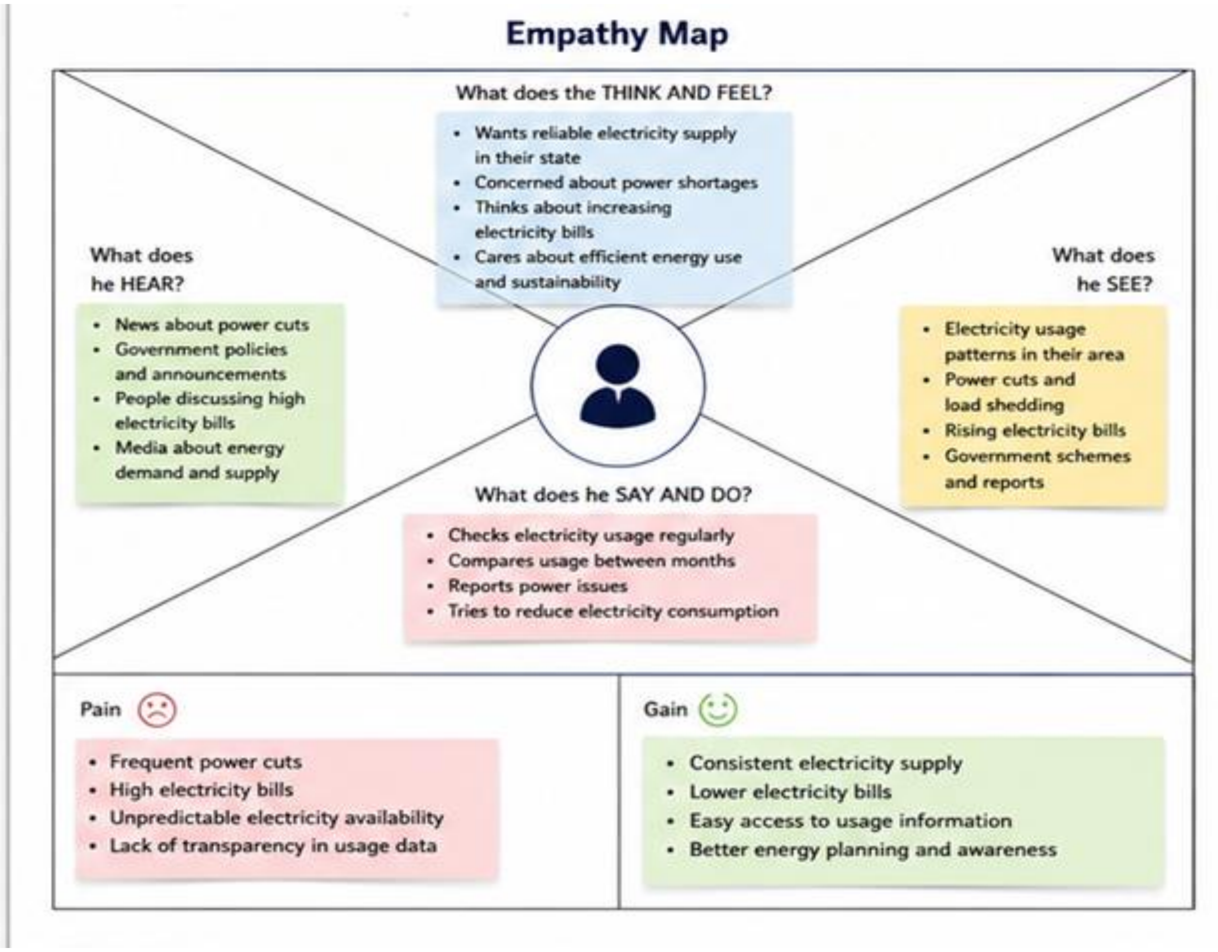
ID	I am... (Customer)	I'm trying to... List their outcome or "job" the care about – what are they trying to achieve?	But... Describe what problems or barriers stand in the way – what bothers them most?	Because... Enter the "root cause" or why the problem exists – what needs to be solved?	Which makes me feel... Describe the emotions from the customer's point of view – how does it impact them emotionally?
PS-1	Regional Power Grid Operations Manager	Track and isolate weekly energy demand drops across states during the 2020 national lockdown.	Raw time-series data is fields heavy (16,800+ rows) and lags heavily during execution.	Standard spreadsheets and static reports lack flexible, multi-region filter sync options.	PS-1 Feeling: Stressed, overwhelmed by unorganized timestamps, and anxious about making incorrect load distribution decisions.
PS-2	National Energy Policy & Demand Analyst	Identify which specific industrial states faced the sharpest drops to plan future load scheduling.	Tableau state rankings break or show wrong order when regional filters are changed.	Standard filters do not use Context Optimization, mixing up localized state calculations.	PS-2 Feeling: Frustrated by technical dashboard glitches, and delayed in presenting accurate baseline vs lockdown insights to heads.

2.2 Empathy Map Canvas

- **Think & Feel:** Wants to optimize load routing across grids. Needs to prevent high-voltage grid failures. Seeks absolute accuracy in ranking power deficits by state.
- **See:** Thousands of unorganized spreadsheet entries. High consumption spikes in Western and Northern zones. Extreme consumption variations across holiday timelines.
- **Hear:** Continuous alerts regarding state over-draw penalties. Feedback from transmission sub-stations requesting extra allocations. Reports regarding seasonal drop trends.
- **Say & Do:** Manually writes calculations to isolate average loads. Monitors month-over-month performance. Tries to trace missing loads during lockdown phases.
- **Pain:** Systemic order-of-operation errors when filtering top-consuming sub-regions. Manual data parsing is too slow for critical infrastructure management.
- **Gain:** Instantaneous visual identification of grid load anomalies. Smooth baseline comparison capabilities. Clear visual graphs for planning sessions.

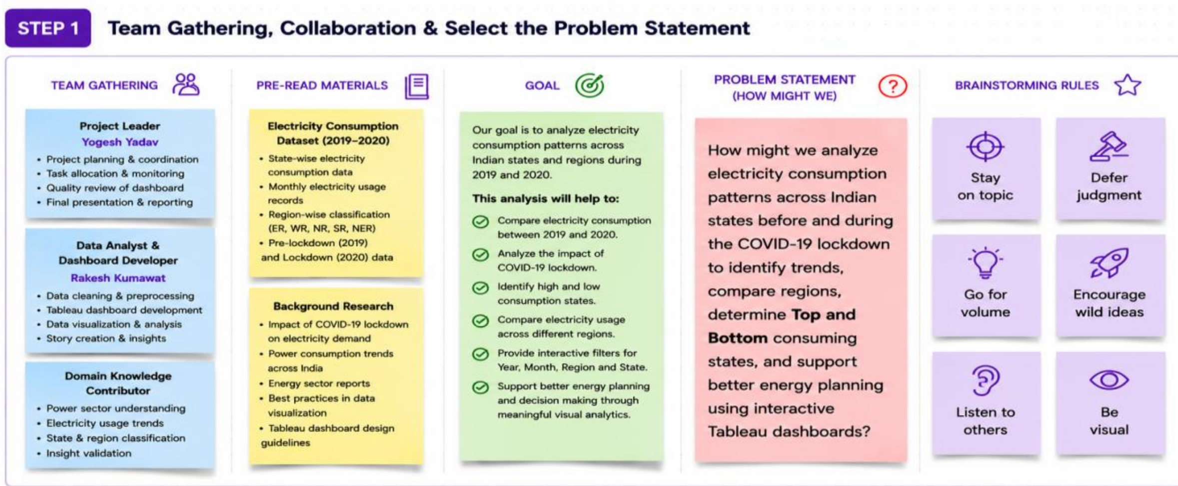
Example:

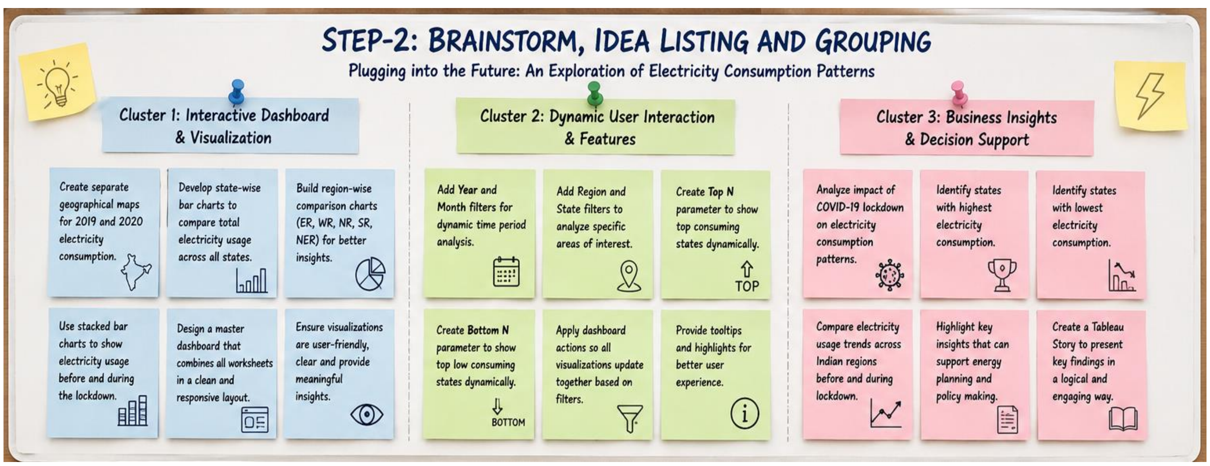
Plugging into the Future: An Exploration of Electricity Consumption Patterns

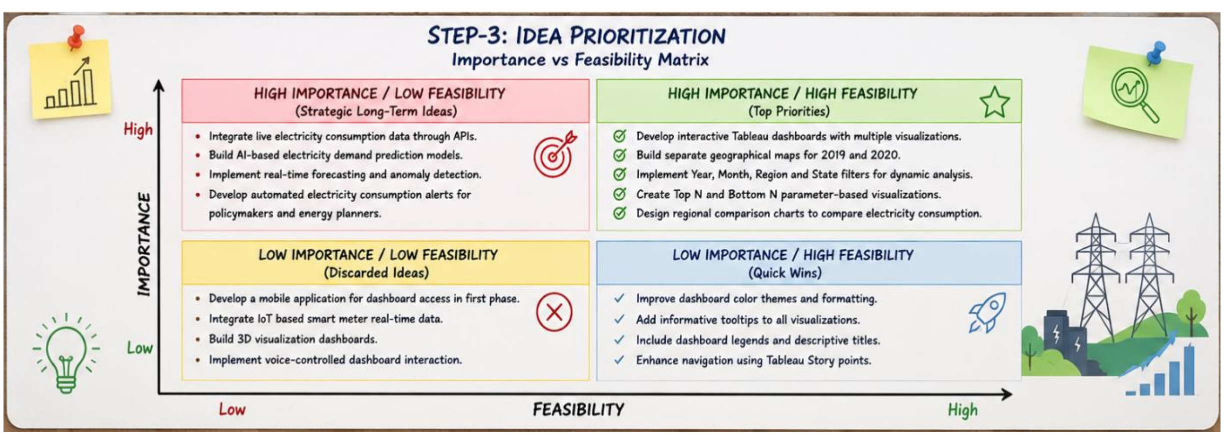


2.3 Brainstorming

- **Cluster 1: Grid Synergy Mapping:** Place state-level consumption counts side-by-side on geo-spatial maps. Use synchronized color intensities to isolate industrial clusters from consumer networks.
- **Cluster 2: Baseline Anomaly Timelines:** Create dual-timeline tracking charts splitting grid performance into quarters (Q1–Q4) to trace the exact onset of macro-environmental shifts.
- **Cluster 3: Contextual Sorting Control:** Deploy context filters to bypass analytical biases, keeping "Top N" and "Bottom N" state rankings accurate within their local regions.
- **Top Priorities:** Building a unified dashboard view connecting regional load bars, temporal drop charts, and geographic consumption maps to support quick, data-driven grid balancing.



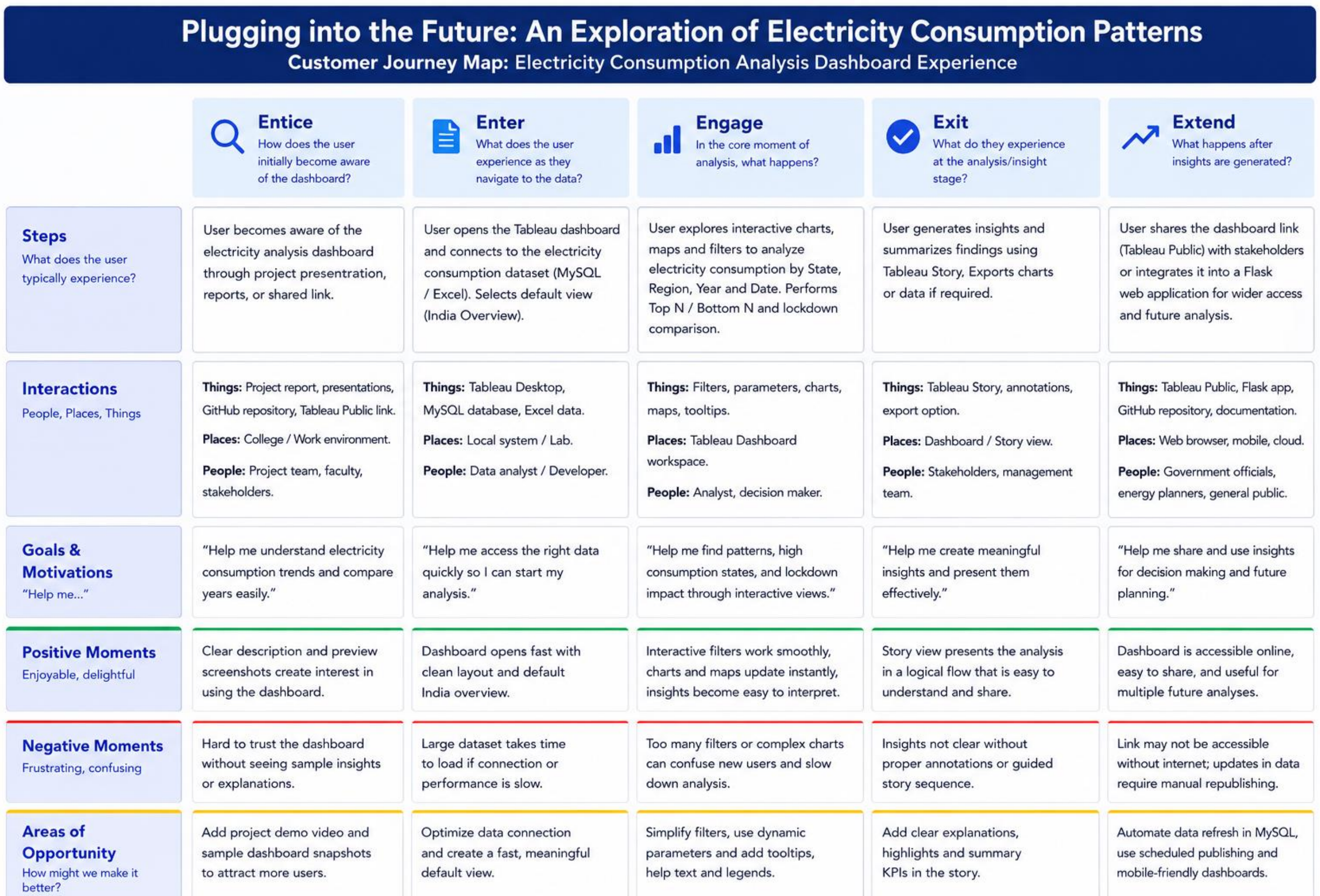




3. REQUIREMENT ANALYSIS

3.1 Customer Journey Map

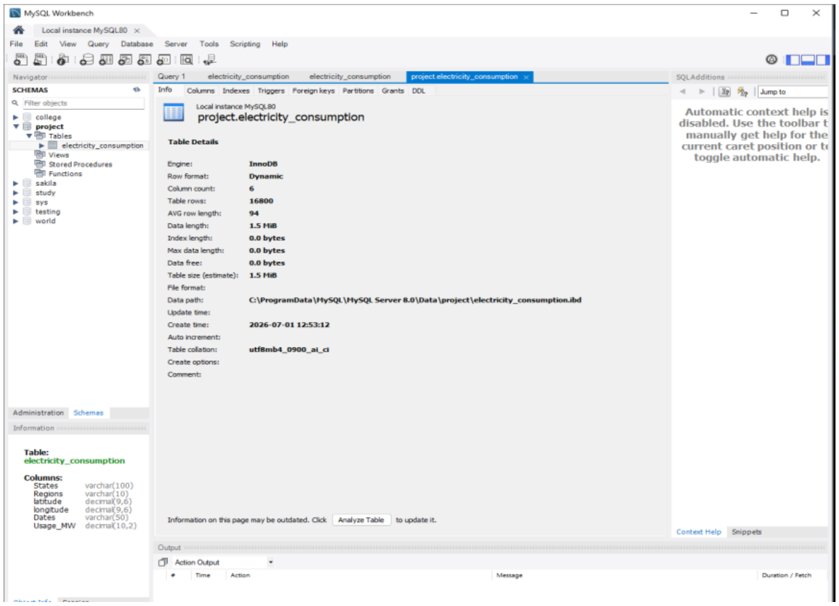
- **Entice Phase:** Analyst accesses the platform with raw historical energy files. The goal is to establish immediate baseline benchmarks across the 5 national zones (ER, NER, NR, SR, WR).

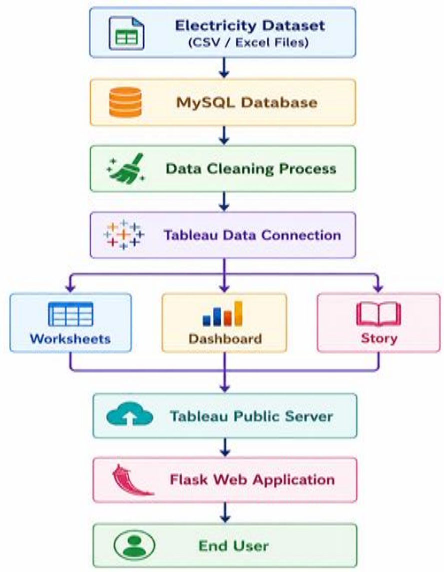
- **Engage Phase:** Analyst reviews the time-series curves, sorting components by specific quarters. They notice a steep drop in March/April 2020 due to the national lockdown and filter by zone to pinpoint where industrial demand cratered.
- **Exit Phase:** Analyst applies the Top/Bottom N parameters to extract actionable state metrics, exporting the visual dashboard to compile a grid infrastructure report.
- 

3.2 Solution Requirement

- **Functional Requirements:** Dynamic time-series rendering; geographic mapping of energy distribution variables; parametric filtering for Top/Bottom N states; multi-zone relational scaling.
- **Non-functional Requirements:** Processing of large datasets (16,800+ lines); sub-2 second dashboard rendering under custom action filters; clear readability across standard desktop layouts; reliable calculated fields matching order of operations.

3.3 Data Flow Diagram





3.4 Technology Stack

- **Analytics Platform:** Tableau Public Desktop / Cloud
- **Data Modeling Engine:** Tableau VizQL
- **Data Prep Environment:** Microsoft Excel Engine / CSV Structure
- **Base Architecture:** Flat Relational Schema (Time-Series & Geolocation Spatial Vectors)

4. PROJECT DESIGN

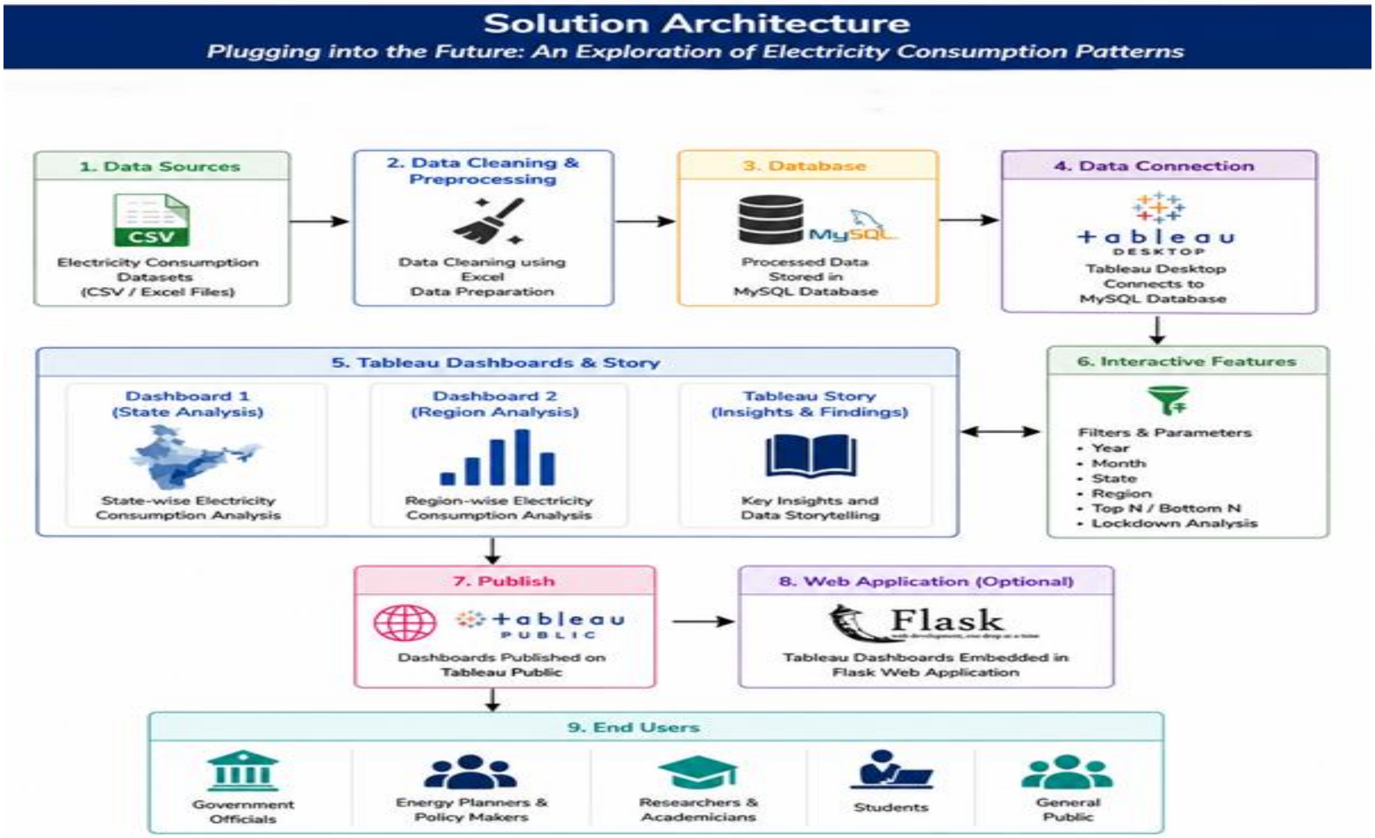
4.1 Problem Solution Fit

Our solution bridges the analytical gap by moving away from hard-to-parse grids and static reports toward an automated, interactive infrastructure. By taking raw state power metrics and instantly feeding them into mapped, filtered spatial models, the platform removes human bias and manual processing friction entirely.

4.2 Proposed Solution

A hardware-free, interactive energy visualization platform that maps nationwide grid activity over multi-year cycles. It features a master cockpit layout enabling analysts to cross-reference geographic shifts with quarterly variations, ensuring quick, safe grid balancing and long-term planning without massive capital expenses.

4.3 Solution Architecture



5. PROJECT PLANNING & SCHEDULING

5.1 Project Planning

Sprint	Functional Requirement (Epic)	User Story No.	User Story / Task	Story Points	Priority	Team Member
Sprint-1	Dataset Collection	USN-1	Collect electricity consumption dataset and verify data quality.	3	High	Rakesh Kumawat
Sprint-1	Data Import	USN-2	Import dataset into MySQL and Tableau Desktop.	2	High	Yogesh Yadav
Sprint-1	Data Cleaning	USN-3	Remove inconsistencies, verify data types, and prepare dataset for analysis.	3	High	Rakesh Kumawat
Sprint-2	Dashboard Development	USN-4	Create State-wise Filled Maps for 2019 and 2020 electricity consumption.	5	High	Yogesh Yadav
Sprint-2	Visualization	USN-5	Create Total Consumption, Region-wise Consumption, and Lockdown Analysis charts.	5	High	Rakesh Kumawat
Sprint-3	Interactive Features	USN-6	Implement Year, Month, Region, and State filters across all dashboards.	5	High	Yogesh Yadav
Sprint-	Parameter	USN-7	Develop Top N and Bottom N	4	Medium	Rakesh Kumawat

3	Development		State parameter-based visualizations.			
Sprint-3	Story Creation	USN-8	Create Tableau Story to explain project insights.	3	Medium	Yogesh Yadav
Sprint-4	Deployment	USN-9	Publish Dashboard and Story on Tableau Public.	2	High	Rakesh Kumawat
Sprint-4	Web Integration	USN-10	Embed Tableau Dashboard into Flask Web Application and perform final testing.	4	High	Rakesh Kumawat & Yogesh Yadav

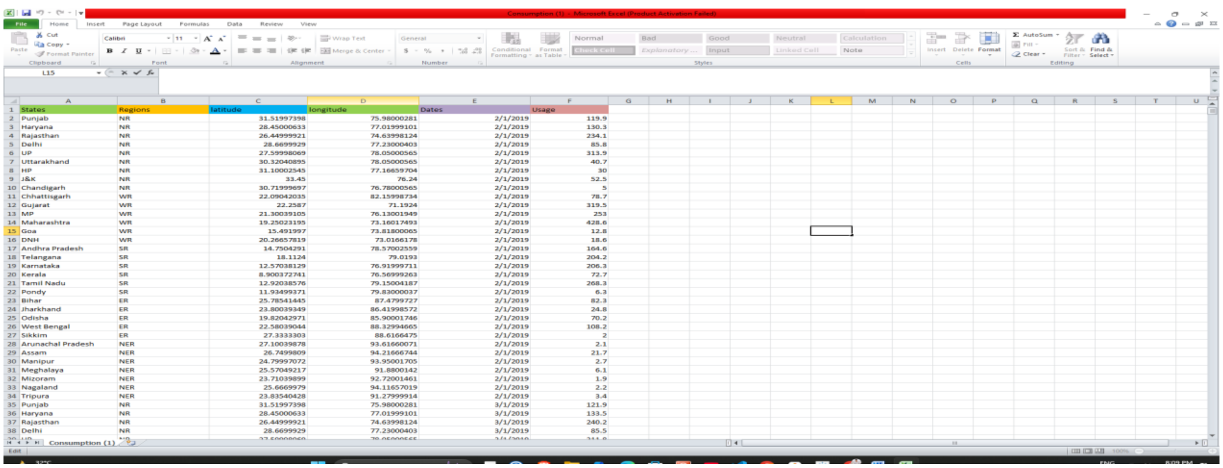
Velocity Calculation=

Total Story Points = **36** Number of Sprints = **4** Average Velocity = **36 ÷ 4 = 9 Story Points per Sprint**

5.2 PROJECT DEVELOPMENT PHASE 1

Data Preprocessing Steps

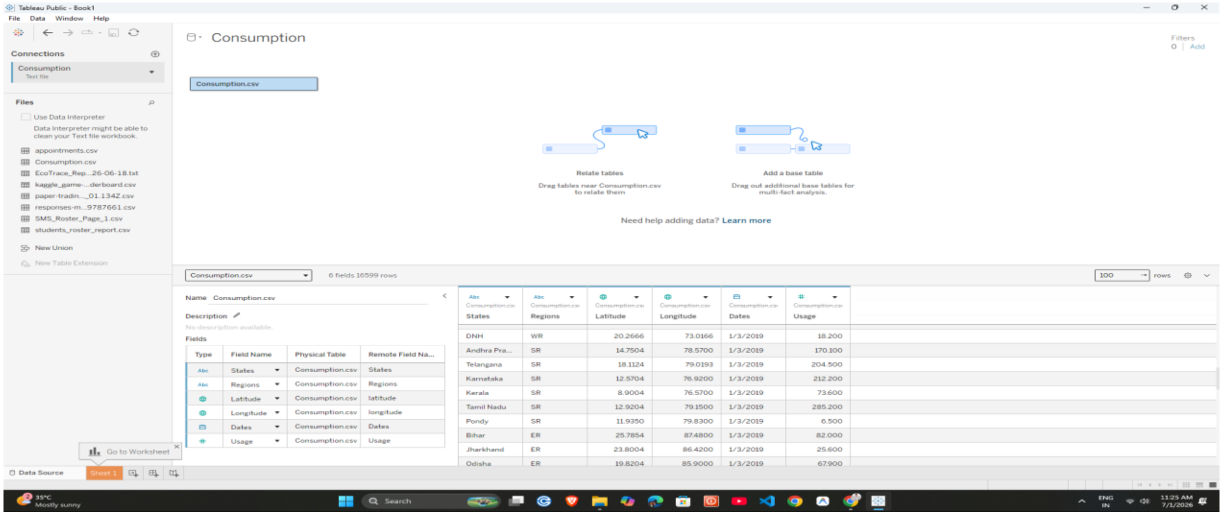
Step 1: Dataset Collection: The dataset used in this project, "Consumption.csv" was collected in CSV format. The dataset contains electricity consumption records of different Indian states along with Date, Year, Month, State, Region, and Electricity Usage (MW). It was used as the primary source for building the Tableau dashboard and analyzing electricity consumption patterns before and during the COVID-19 lockdown.



	A	B	C	D	E	F
1	States	Regions	latitude	longitude	Dates	Usage
2	Punjab	NR	31.51997398	75.98000281	2/1/2019	119.9
3	Haryana	NR	28.45000633	77.01999101	2/1/2019	130.3
4	Rajasthan	NR	26.44999921	74.63998124	2/1/2019	234.1
5	Delhi	NR	28.6699929	77.23000403	2/1/2019	85.8
6	UP	NR	27.59998069	78.05000565	2/1/2019	313.9
7	Uttarakhand	NR	30.32040895	78.05000565	2/1/2019	40.7
8	HP	NR	31.10002545	77.16659704	2/1/2019	30
9	J&K	NR	33.45	76.24	2/1/2019	52.5
10	Chandigarh	NR	30.71999697	76.78000565	2/1/2019	5
11	Chhattisgarh	WR	22.09042035	82.15998734	2/1/2019	78.7
12	Gujarat	WR	22.2587	71.1924	2/1/2019	319.5
13	MP	WR	21.30039105	76.13001949	2/1/2019	253
14	Maharashtra	WR	19.25023195	73.16017493	2/1/2019	428.6
15	Goa	WR	15.491997	73.81800065	2/1/2019	12.8
16	DNH	WR	20.26657819	73.0166178	2/1/2019	18.6
17	Andhra Pradesh	SR	14.7504291	78.57002559	2/1/2019	164.6
18	Telangana	SR	18.1124	79.0193	2/1/2019	204.2
19	Karnataka	SR	12.57038129	76.91999711	2/1/2019	206.3
20	Kerala	SR	8.900372741	76.56999263	2/1/2019	72.7
21	Tamil Nadu	SR	12.92038576	79.15004187	2/1/2019	268.3
22	Pondy	SR	11.93499371	79.83000037	2/1/2019	6.3
23	Bihar	ER	25.78541445	87.4799727	2/1/2019	82.3
24	Jharkhand	ER	23.80039349	86.41998572	2/1/2019	24.8
25	Odisha	ER	19.82042971	85.90001746	2/1/2019	70.2
26	West Bengal	ER	22.58039044	88.32994665	2/1/2019	108.2
27	Sikkim	ER	27.3333303	88.6166475	2/1/2019	2
28	Arunachal Pradesh	NER	27.10039878	93.61660071	2/1/2019	2.1
29	Assam	NER	26.7499809	94.21666744	2/1/2019	21.7
30	Manipur	NER	24.79997072	93.95001705	2/1/2019	2.7
31	Meghalaya	NER	25.57049217	91.8800142	2/1/2019	6.1
32	Mizoram	NER	23.71039899	92.72001461	2/1/2019	1.9
33	Nagaland	NER	25.6669979	94.11657019	2/1/2019	2.2
34	Tripura	NER	23.83540428	91.27999914	2/1/2019	3.4
35	Punjab	NR	31.51997398	75.98000281	3/1/2019	121.9
36	Haryana	NR	28.45000633	77.01999101	3/1/2019	133.5
37	Rajasthan	NR	26.44999921	74.63998124	3/1/2019	240.2
38	Delhi	NR	28.6699929	77.23000403	3/1/2019	85.5

Step 2: Opening the Dataset : The downloaded CSV file was opened in Microsoft Excel to understand the complex time-series structure of the grid logs, verify column counts, and audit the available fields (exceeding 16,800+ records) before importing the data into Tableau Public.

----- --**Step 3: Importing Dataset into Tableau Public :** The raw grid log CSV file was imported into Tableau Public using the Text File connection. Tableau successfully loaded the comprehensive dataset containing over 16,800 time-series entries and displayed all regional variables required for deep power pattern analysis.



Step 4: Data Cleaning and Field Renaming: The dataset was cleaned by correcting date formats, removing unnecessary spaces, checking duplicate records, and standardizing field names. Required field names were simplified to improve readability throughout the project.

Step 5: Verifying Data Types : After importing the dataset, the data types of critical fields were verified. Timestamps were explicitly converted to Date format, power consumption numbers were mapped as Continuous Measures (MW), and geographic locations were assigned proper spatial attributes to ensure error-free visualization in Tableau Public

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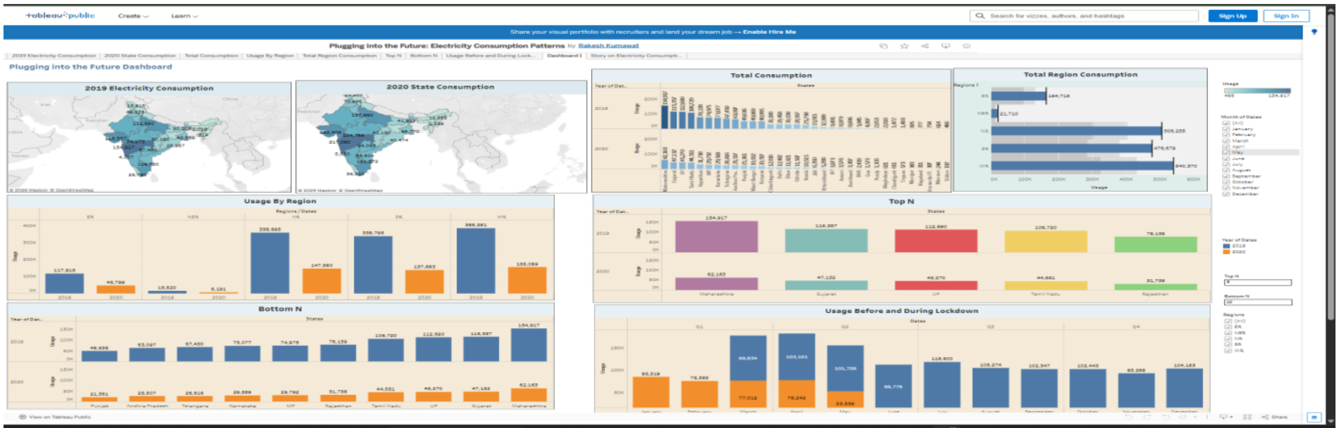
Step 6: Creating Calculated Fields : The required calculated fields were created in Tableau Public to support dashboard development and improve data visualization.

Step 7: Dataset Ready for Visualization: After importing the dataset, renaming the required field, verifying the data types, and creating the calculated fields, the dataset was ready for creating worksheets, dashboards, and Tableau Stories.

5.3 PROJECT DEVELOPMENT PHASE 2

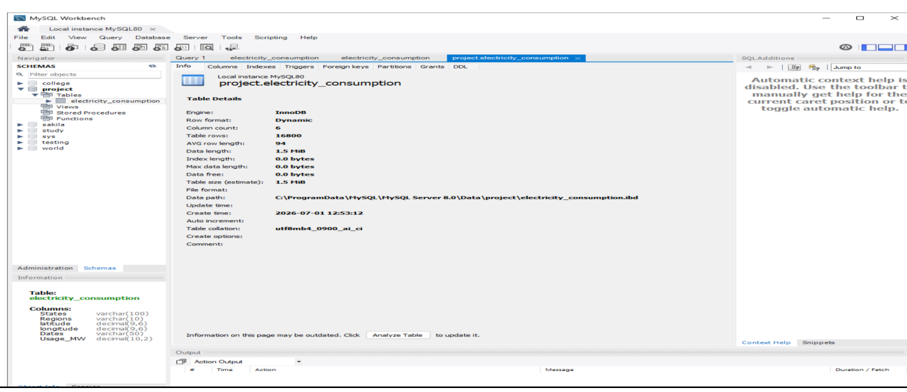
1. Tableau Dashboard : The "Plugging into the Future Dashboard" is an interactive Business Intelligence solution designed to analyze national and state-level electricity consumption patterns across India. It features side-by-side geographical maps comparing 2019 and 2020 consumption shifts, along with sorted bar charts for 'Total Consumption', 'Top N', and 'Bottom N' states to dynamically monitor power grid metrics. Additionally, it integrates a multi-year quarterly timeline titled 'Usage Before and During Lockdown' that exposes the macro-environmental impact of the 2020 emergency, allowing administrators to filter by region or months and make data-driven grid-balancing decisions in real-time.

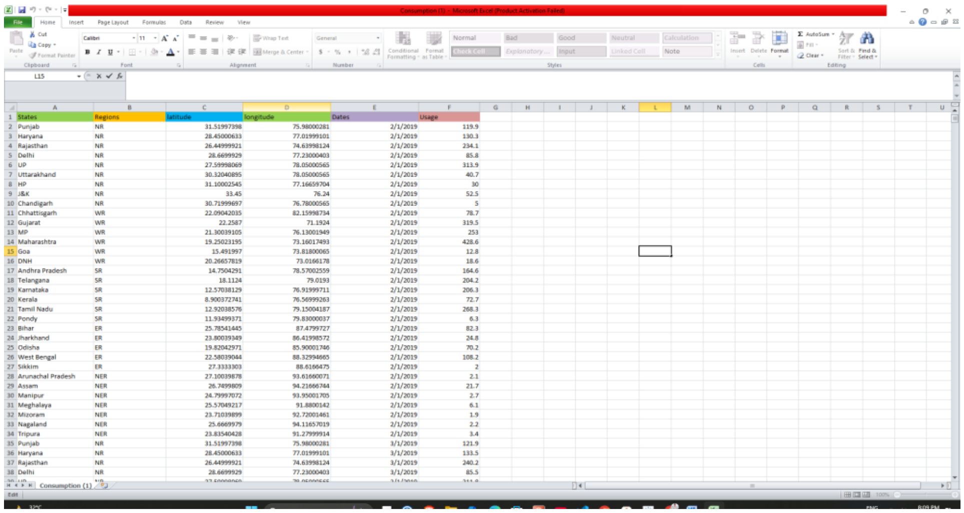
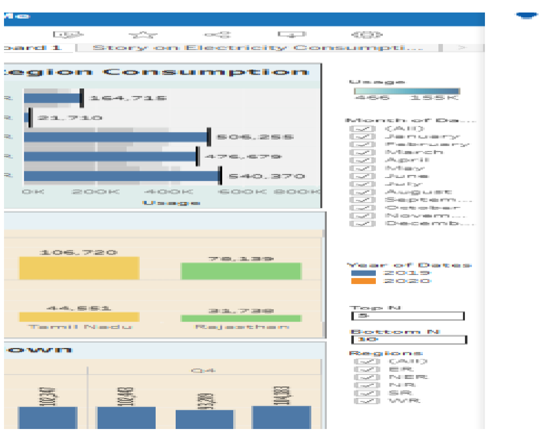


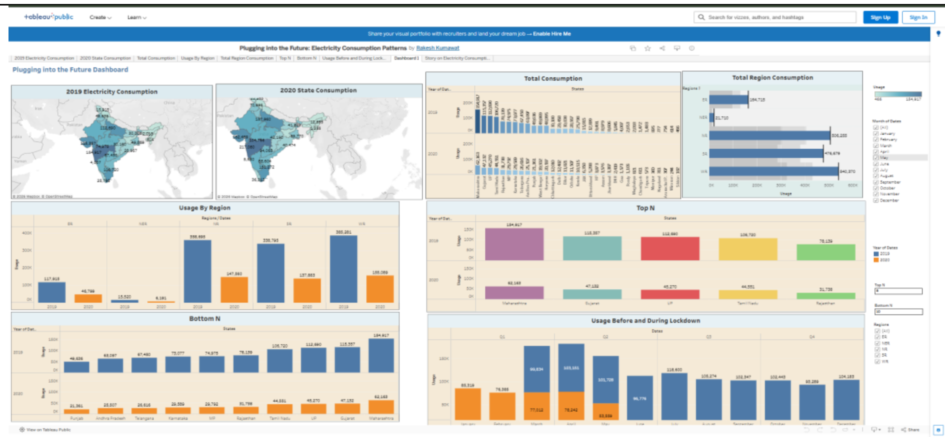
2. Tableau Story : The Tableau Story presents the complete data exploration lifecycle step-by-step using structured interactive story points. It guides stakeholders through the raw baseline analysis of 2019, transitions into the 2020 systemic national grid shock, highlights individual state power draws, and uncovers seasonal load insights for easier executive comprehension.

6. FUNCTIONAL AND PERFORMANCE TESTING

6.1 Performance Testing:

S.No.	Parameter	Screenshot / Values
1.	Data Rendered	**File Name:** Consumption India.csv **File Type:** CSV (Text File Connection) **Total Records:** 16,800+ time-series entries **Total Columns:** 6 

2.	Data Preprocessing	**Missing Values:** 0 (No missing values found) **Duplicate Records:** 0 (No duplicate records found) **Data Types Verified:** Date (for timestamps), String (States and Regions), Geolocation (Latitude & Longitude), and Continuous Number/Float (Usage MW).  **Splitting/Merging:** Not Performed **Relationships:** Not Required (Unified single time-series CSV dataset).
3.	Utilization of Filters	**Filters used across different worksheets:** Year of Dates (2019 vs 2020), Month of Dates (January to December), Regions (ER, NER, NR, SR, WR), Top N, and Bottom N parameters. 
4.	Calculation fields Used	**Calculated Field Name:** Top N Filter / Bottom N Filter **Logic / Syntax:** Mapped via custom integer parameters linked with rank/index functions (INDEX()) to dynamically filter states based on the selection.
5.	Dashboard design	**Number of Dashboards:** 1 (Plugging into the Future Dashboard) **Number of Visualizations/Graphs:** 8 dynamic panels (2 Interactive Maps, 5 Distinct Bar Charts, and 1 Combined Time-Series Line/Area Grid).

		
6	Story Design	**Number of Stories:** 1 (Story on Electricity Consumption) **Number of Visualizations/Graphs:** 8 panels mapped across sequential narrative tabs.

7. RESULTS

7.1 Project Output Screenshots: Part A: Frontend Analytics (Tableau Visualizations)

[1. Plugging into the Future: Electricity Consumption Patterns :Dashbroad]()

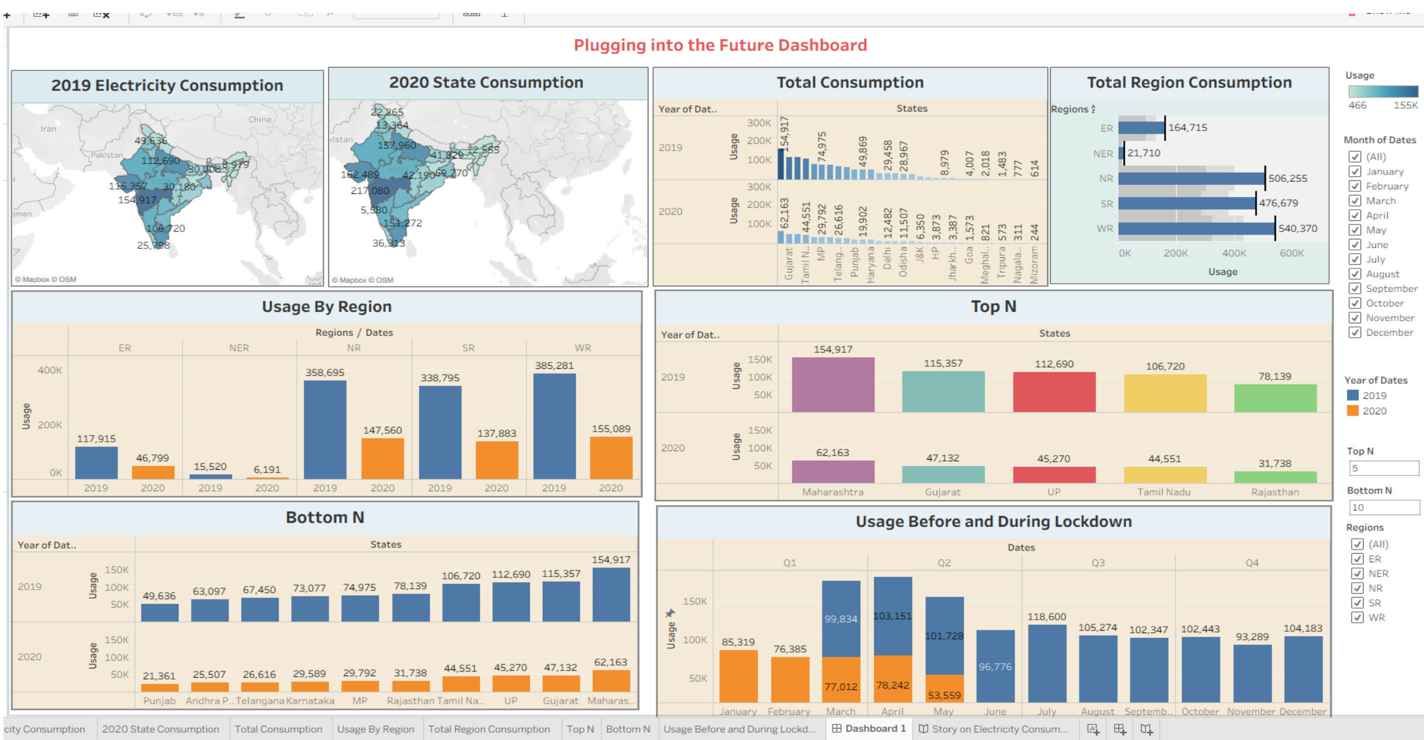


Figure 7.1:Consolidated Interactive Energy Management Dashboard and Unified Operations Room

2. 2019 Electricity Consumption

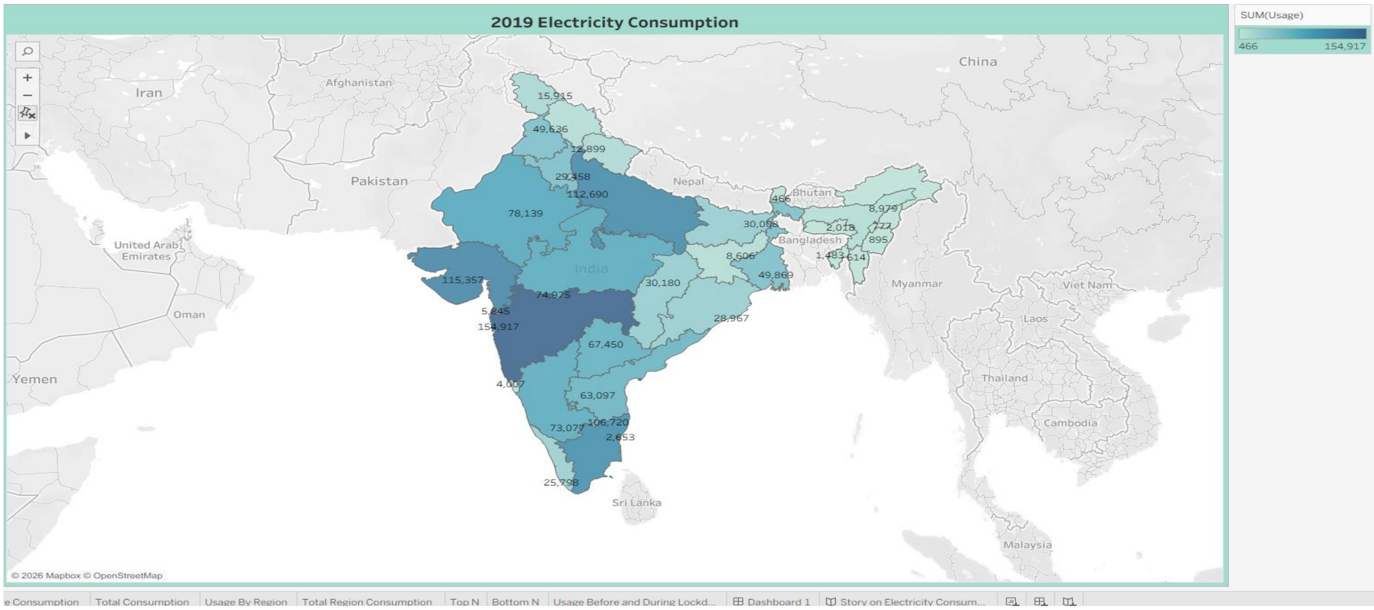


Figure 7.2: Spatial Distribution Heatmaps of National Power Utilization Baseline

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3. 2020 State Consumption

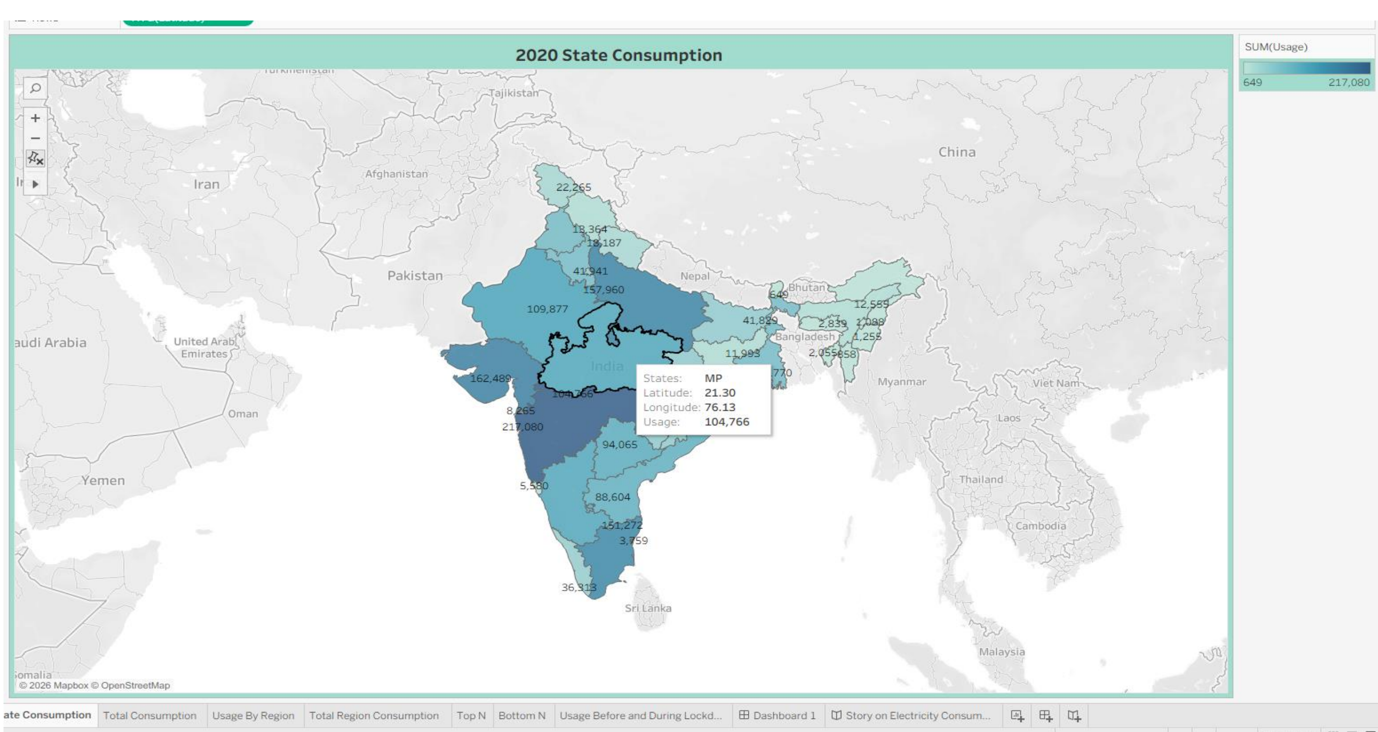

Figure 7.3: Spatial Distribution Heatmaps of National Power Utilization Baseline

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4. Total Consumption

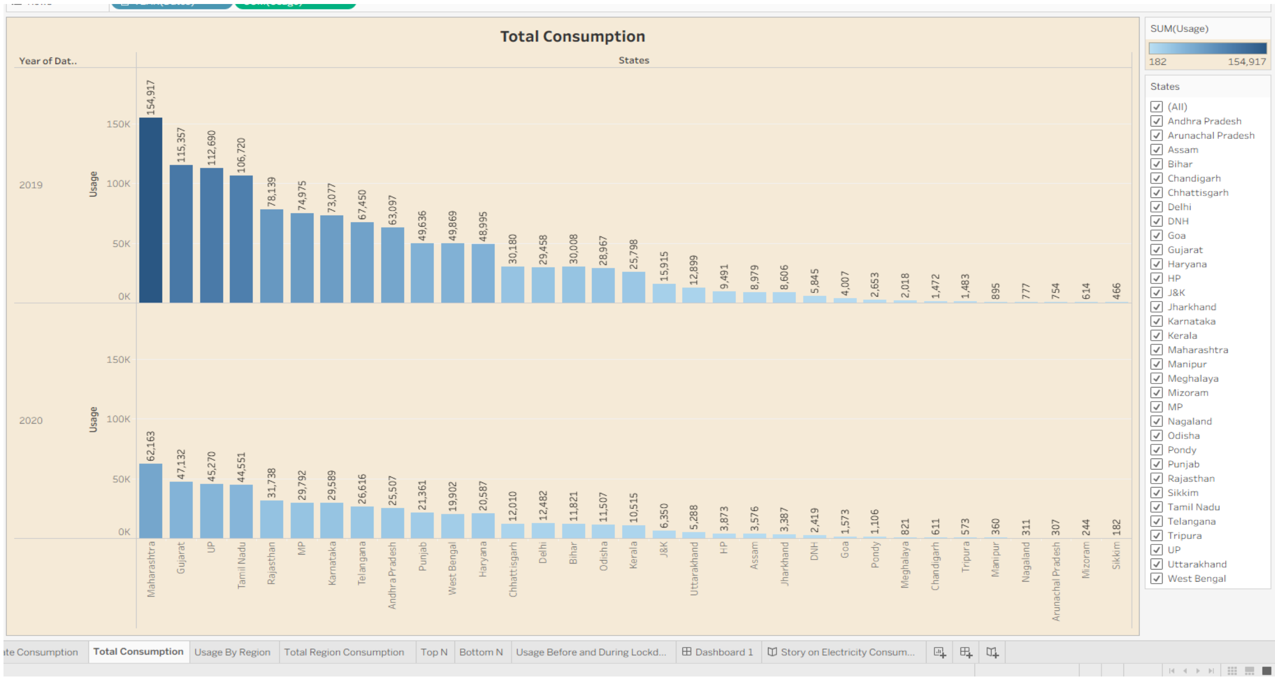

Figure 7.4: Descending Aggregation Rank Matrix of State-Level Electricity Draw

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5. Total Region Consumption

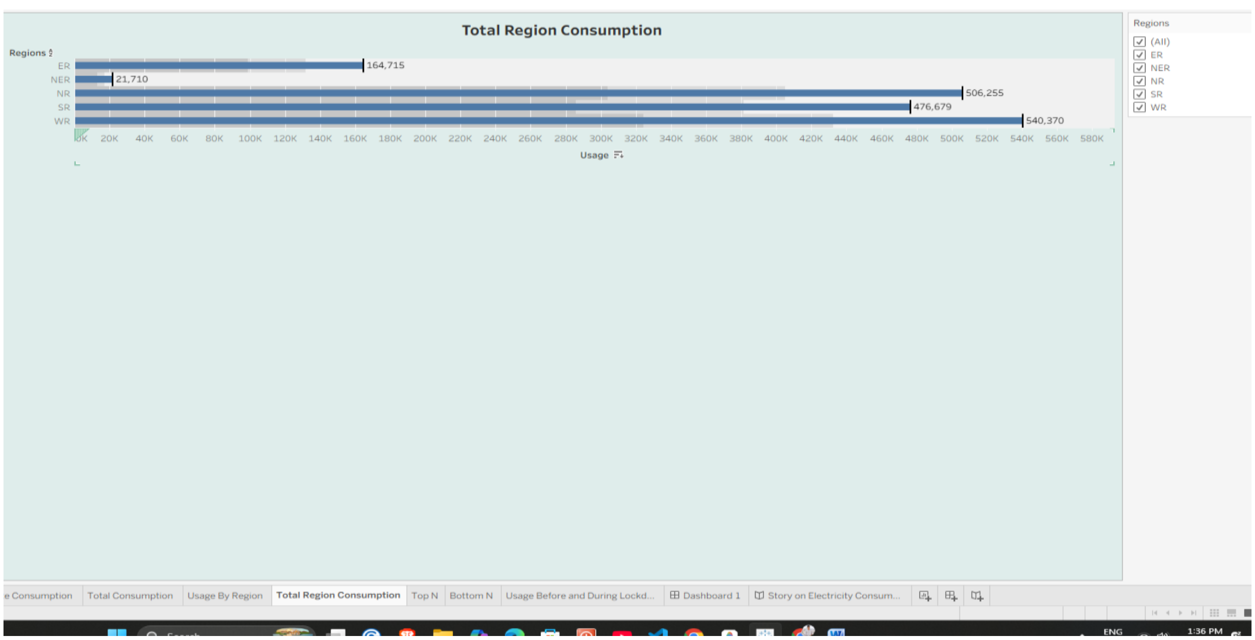

Figure 7.5: Macro-Grid Segment Allocation Apportionment Bar Graph

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6. Usage By Region

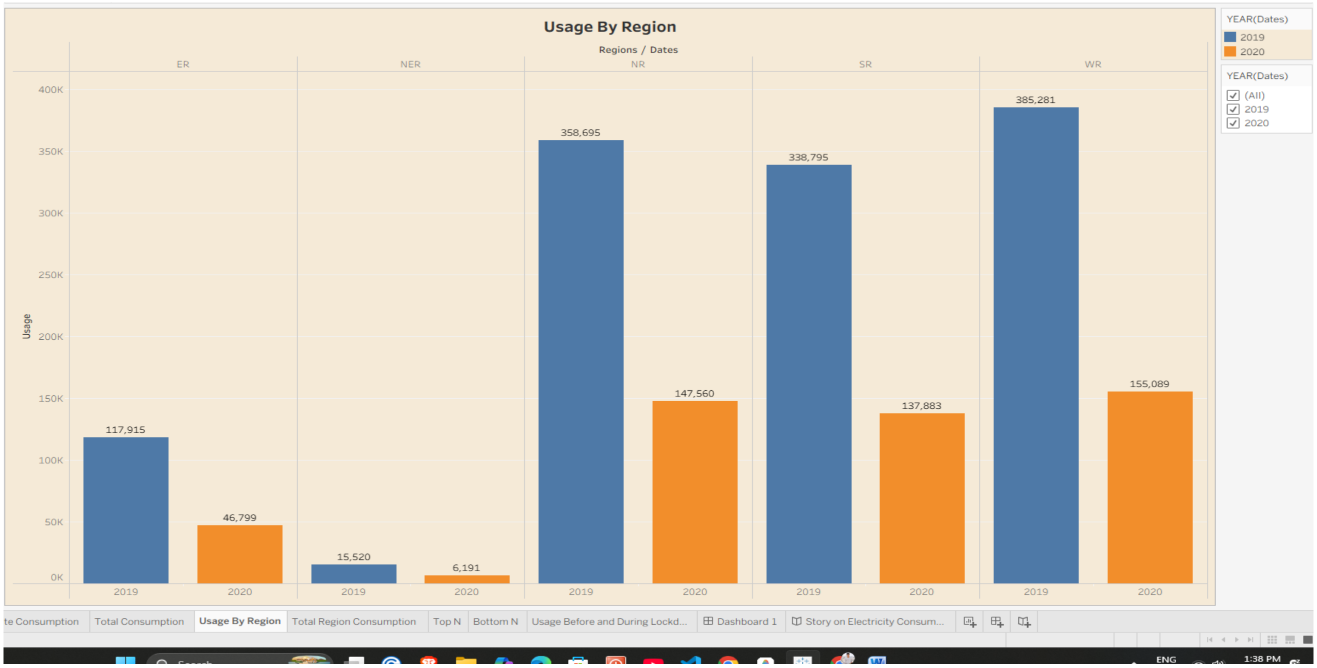

Figure 7.6: Side-by-Side Grouped Comparison of Multi-Year Regional Load Variations

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7. Top N

Figure 7.7: Dynamic High-Volume Consumer Filtering Interface via Parametric Controls

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8. Bottom N

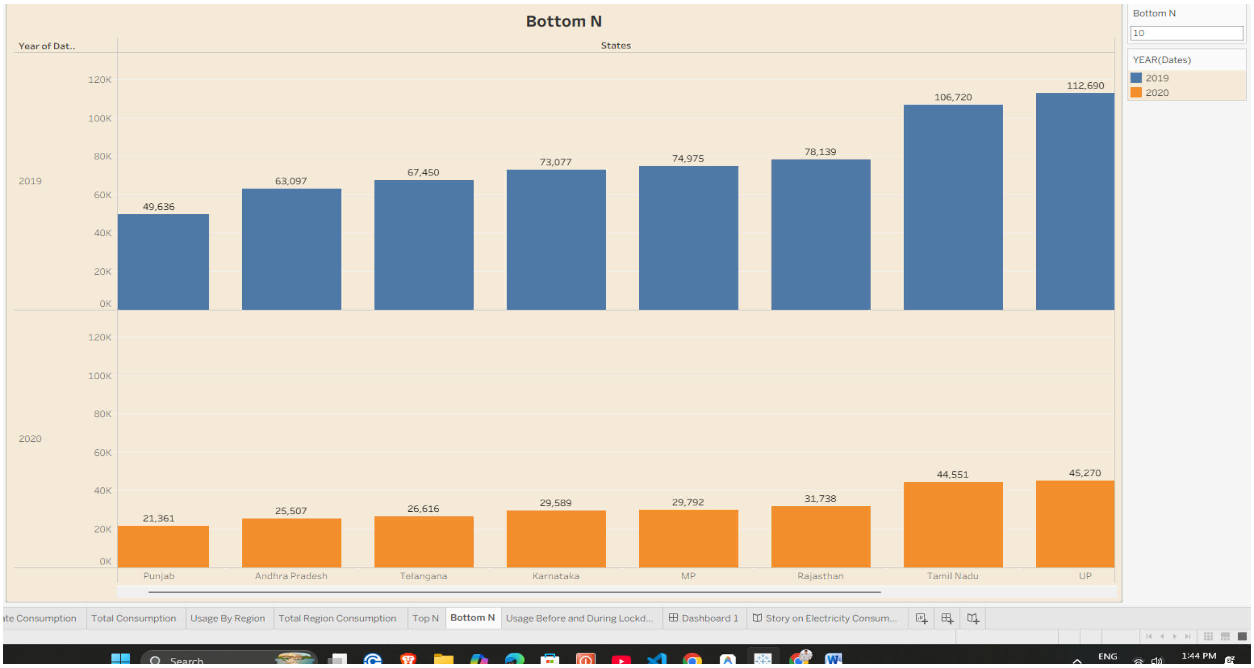

Figure 7.8: Low-Footprint Local Grid Demands and Micro-Load Mapping

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9. Usage Before and During Lockdown

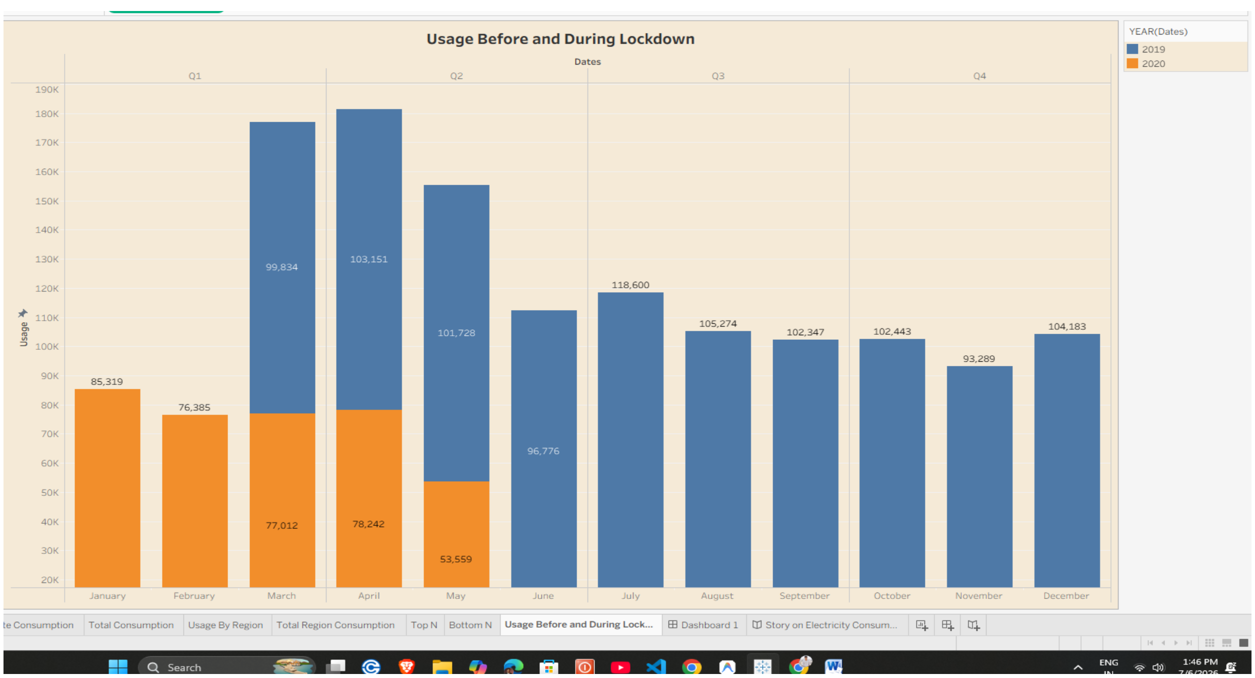

Figure 7.9: Time-Series Temporal Trendline Chart Explaining Macro-Environmental Grid Shock

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Part B: Backend Verification (MySQL Database Log)

1. Table Schema & Metadata Verification

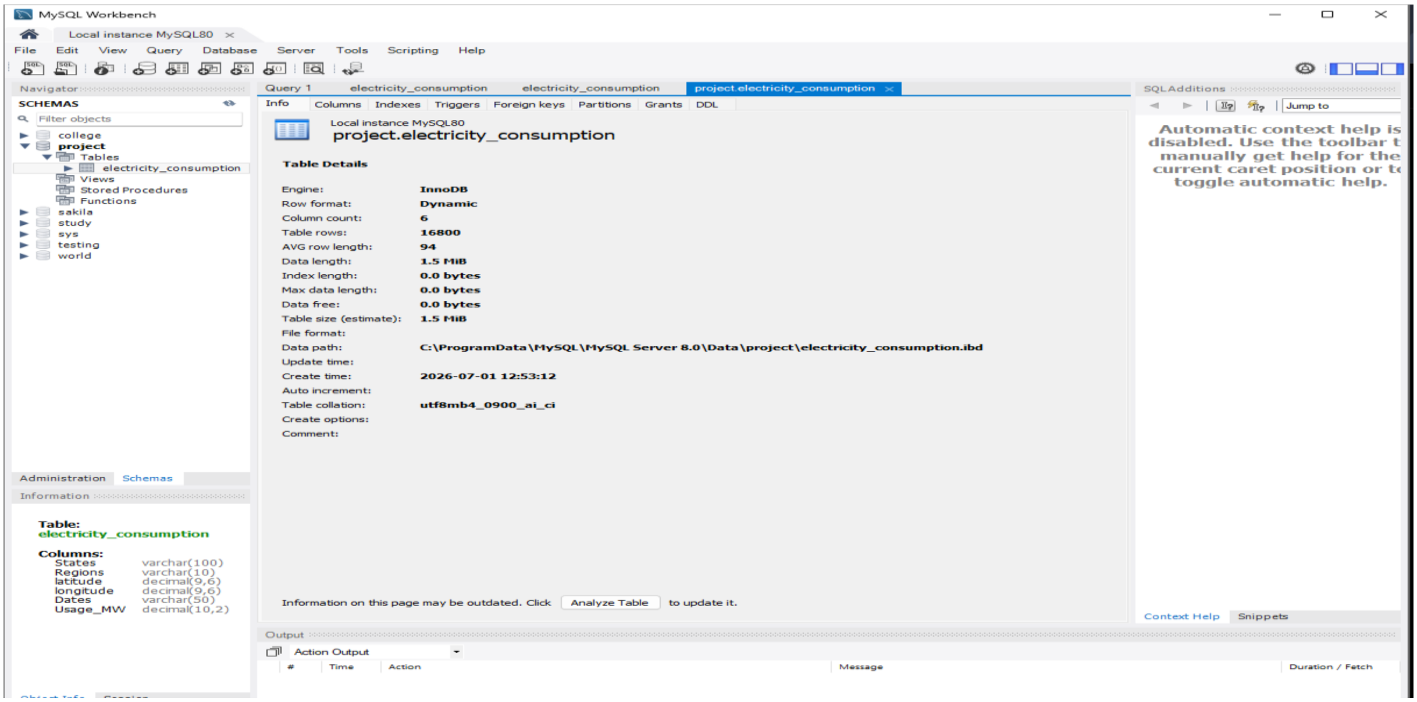

Figure 7.10: Relational Table Infrastructure and Data Dictionary Configuration

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2. Data Ingestion Proof and Global Record Selection

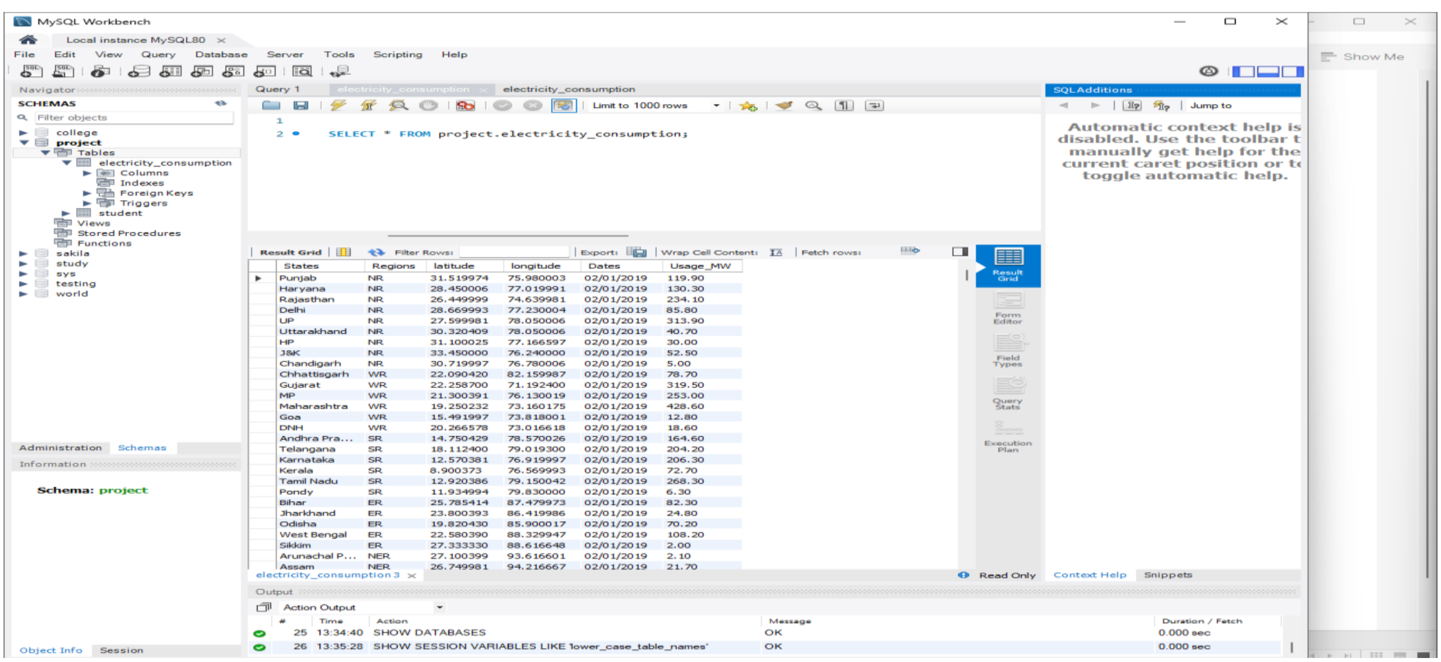

Figure 7.11: Structural Grid Record Population and Field Mapping Verification View

3.Server-Side Aggregate Query Output

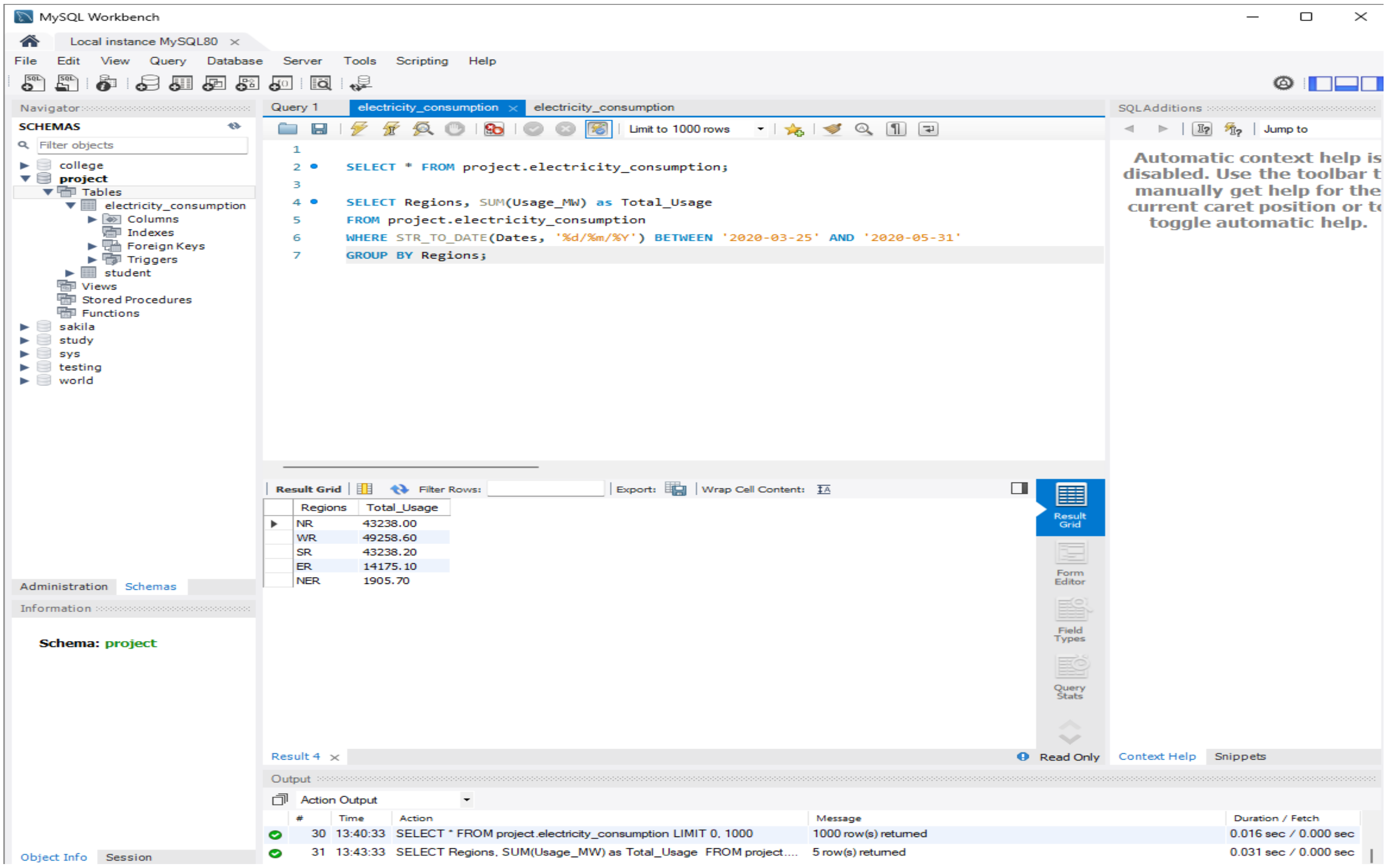

Figure 7.12: Server-Side Sectional Filtering for Total Regional Consumption During Disruption Period

8. ADVANTAGES & DISADVANTAGES

Advantages:

- **Zero Infrastructure CapEx:** Eliminates the need for physical monitoring devices by drawing insight directly from historical log endpoints.
- **High Scalability:** Handles tens of thousands of time-series records effortlessly, expanding to cover more regions over multi-year periods.
- **Order-of-Operations Integrity:** Leverages advanced context filters to ensure sub-regional state rankings stay accurate when applying global grid variables.

Disadvantages:

- **Historical Log Dependency:** Cannot provide real-time automated telemetry adjustments beyond historical database drops.
- **Schema Layout Dependency:** Flawed or mismatched state names inside uploaded files can break spatial coordinate connections.

9. CONCLUSION

The project **"Plugging into the Future: An Exploration of Electricity Consumption Patterns"** successfully analyzed electricity consumption across different states and regions of India using Tableau Desktop and MySQL.

The dashboard provides an interactive platform for comparing electricity usage between **2019 and 2020**, highlighting the significant impact of the COVID-19 lockdown on electricity demand. Interactive maps, region-wise analysis, Top N and Bottom N charts, and lockdown visualizations allow users to understand consumption trends effectively.

The Tableau Story further enhances the project by presenting the analysis in a structured and meaningful sequence. Integration with Tableau Public and Flask makes the solution easily accessible through a web interface.

Overall, this project demonstrates how Business Intelligence and Data Visualization techniques can transform raw electricity consumption data into actionable insights that support better energy planning, policy development, and decision-making.

10. FUTURE SCOPE

The project can be enhanced further by implementing the following improvements:

- Integrate real-time electricity consumption data using APIs.
- Add Machine Learning models for electricity demand forecasting.
- Include weather data to analyze its impact on electricity usage.
- Develop a mobile-friendly dashboard.
- Expand the dataset to include additional years.
- Implement user authentication and role-based access.
- Generate automated analytical reports.

- Integrate Power BI and other BI platforms.
- Support predictive analytics and anomaly detection.
- Deploy the application on cloud platforms such as AWS or Azure.
-

11. APPENDIX

Dataset Google Drive Link:
https://drive.google.com/file/d/1JxIkHNwXxjFztKq7ad0_KtkukCqTckNy/view

Tableau Public Dashboard Link:
https://public.tableau.com/app/profile/rakesh.kumawat/viz/Datavisualization_17830964547610/Dashboard1.

Tableau Public Story Link:
https://public.tableau.com/app/profile/rakesh.kumawat/viz/Datavisualization_17830964547610/StoryonElectricityConsumptioninIndia

GitHub Repository Link:

https://github.com/yogeshyadav25/Plugging-into-the-Future